

Eye Injuries

Key Concepts

- Eye injuries are often overlooked in the trauma patient.
- 18.5% of patients with facial fractures will have associated eye injuries.
- Teardrop pupils, pain, vision problems, difficulty differentiating color, and blood in eye may signify an eye injury.
- The key goal of care is to prevent further damage to the injured eye.



Major vs Minor Eye Injuries

Major

- Penetrating injury
- Blunt trauma with large, heavy, or high velocity object
- Alkali burns
- Hydrofluoric acid burns
- Eye injury associated with other significant traumatic injury or altered mental status
- Eye injury associated with true loss of vision

Minor

- Corneal scratch or abrasion
- Minor chemical burns
- UV burns – welding, sun exposure

History

- What caused injury? – Blunt, penetrating, chemical, burn
- Chemical – see Eye Injury Chemical Guideline
- Burn – Thermal, UV (welding)?
- Other associated injuries? – Head injury, facial injury, spinal injury, altered LOC, etc.?
- Loss of vision?
- Did patient wash out eye?
- Other care given prior to EMS arrival?
- Other significant medical history?

Physical Exam

- Primary and Secondary Trauma Exam
- Assess for obvious trauma to eye or penetrating object
- Visual acuity exam in each eye (if patient can tolerate)
- Look for pupil irregularity, blood in anterior chamber of eye, fluid leaking from orbit
- Note other traumatic injury – eyelid lacerations, bruising, etc.
- Mental status assessment

EMT/AEMT

Shield any suspected penetrating or blunt force eye injury.

- Do not place anything (gauze, foam, etc) on the eye.
- Place metal eye shield over the eye. The shield should only touch orbital bones
- Do not attempt to remove any foreign bodies from the eye.
- Consider shielding both eyes in the setting of retained foreign body.
- Styrofoam cups can be taped over the eye if a shield is not available.

EMT/AEMT - Expanded Scope

- **Zofran** 4mg ODT or IM, may repeat once in 5 min (max 8mg)
- Pain management Guideline

Paramedic

- Elevate head of bed to 30°.
- Administer **Zofran** 4 mg IV for nausea / vomiting, may repeat once in 5 min (max 8mg)
- Pain Management guideline